

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 32 — Hydroxy Compounds (Phenol)

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Structure & Acidity of Phenols · Delocalisation in Phenoxide · Electrophilic Substitution · Halogenation & Nitration
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**32.1 Structure & Relative Acidity of Phenol**

— Comparison of acid strengths of ethanol, water, phenol, and ethanoic acid; delocalisation of negative charge in phenoxide

32.2 Ring Activation & Electrophilic Substitution— Overlap of the oxygen p-orbital lone pair with the delocalised benzene π -system; activation of 2-substituted phenols**32.3 Chemical Tests & Distinction from Alcohols/Acids**

— Characteristic purple/violet complexation with neutral aqueous iron(III) chloride; rapid decolourisation of bromine water

32.4 Diazonium Coupling & Azo Dyes— Electrophilic aromatic substitution of alkaline phenol by the benzenediazonium cation ($\text{C}_6\text{H}_5\text{N}_2^+$)**High-Frequency Core Repeats (Q41–Q50)**

— The 10 most frequently tested Cambridge Paper 4 questions on Hydroxy Compounds and Phenols

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Relative Acidities of Ethanol, Water, and Phenol — 9701/42/M/J/23/Q7 [6 Marks]

HARD · Syllabus 32.1

The relative acid strengths of ethanol, water, and phenol differ significantly as shown in Fig. 1.1.

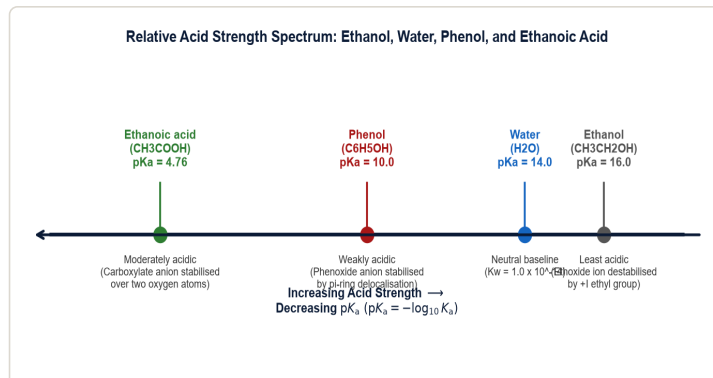


Fig. 1.1: Relative acid strength and pK_a spectrum for ethanol, water, phenol, and ethanoic acid.

- (a) Rank ethanol, water, and phenol in order of increasing acid strength. [1]
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- (b) Explain in terms of structure and bonding why phenol is significantly more acidic than ethanol. [3]
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- (c) Compare the reactions of ethanol and phenol with aqueous sodium hydroxide, NaOH(aq) , writing an ionic equation where a reaction occurs. [2]
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2 Resonance Delocalisation in the Phenoxide Anion and Ring Activation — 9701/41/O/N/22/Q6 [6 Marks]

HARD · Syllabus 32.1

The chemical behaviour of phenol is governed by interaction between the oxygen lone pair and the aromatic ring as shown in Fig. 2.1.

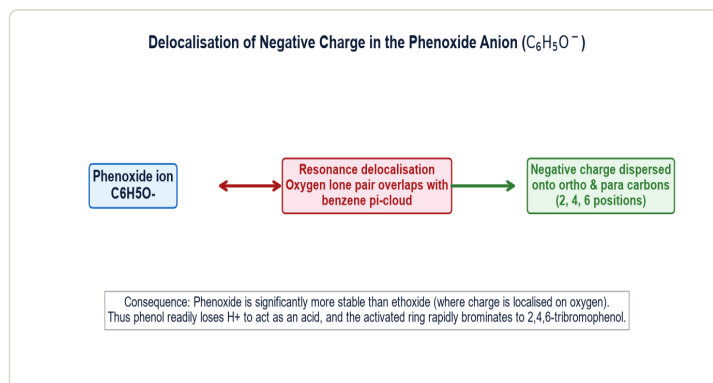


Fig. 2.1: Delocalisation of negative charge across the phenoxide system.

- (a) Describe how the lone pair of electrons on the oxygen atom interacts with the aromatic ring in phenol. [2]
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- (b) State which ring positions in phenol experience the greatest increase in electron density as a consequence. [1]

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- (c) Explain why phenol reacts with dilute nitric acid at room temperature, whereas benzene requires a mixture of concentrated nitric and sulfuric acids at 55 °C. [3]

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3 Bromination of Phenol vs Bromination of Benzene — 9701/42/M/J/22/Q7 [6 Marks]

HARD · Syllabus 32.2

- (a) State the observations when aqueous bromine is added to phenol at room temperature. [2]

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- (b) Give the structural formula and systematic IUPAC name of the organic product formed in (a). [2]

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- (c) Contrast the reaction conditions and mechanism for the bromination of benzene with those for the bromination of phenol. [2]

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4 Distinguishing Tests Between Phenol, Ethanoic Acid, and Cyclohexanol — 9701/41/M/J/21/Q8 [6 Marks]

HARD · Syllabus 32.3

- (a) Describe a chemical test that will give a positive result with ethanoic acid but no reaction with phenol. [2]

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- (b) Describe a chemical test using iron(III) chloride that distinguishes phenol from cyclohexanol. [2]

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- (c) Describe how the reactions of cyclohexanol and phenol with sodium metal compare in observations and products. [2]

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5 Azo Dye Synthesis: Diazotisation and Alkaline Coupling with Phenol — 9701/42/O/N/20/Q6 [6 Marks]

HARD · Syllabus 32.4

- (a) State the reagents and precise temperature required to convert phenylamine into the benzenediazonium ion. [2]

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- (b) Explain why the temperature in (a) must be kept strictly below 10 °C. [1]

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- (c) Describe the conditions required for benzenediazonium chloride to react with phenol, and give the structure of the azo dye produced. [3]

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6 Synthesis and Acidity of Substituted Phenols: Nitrophenols — 9701/43/M/J/23/Q7 [6 Marks]

HARD · Syllabus 32.1

- (a) Explain why 4-nitrophenol is significantly more acidic than unsubstituted phenol. [3]

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- (b) Explain why 4-methylphenol is slightly less acidic than unsubstituted phenol. [2]

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- (c) Predict whether 2,4,6-trinitrophenol (picric acid) will react with aqueous sodium hydrogencarbonate, giving a reason. [1]

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7 Esterification Reactions of Phenol and Ethanoyl Chloride — 9701/42/F/M/22/Q8 [6 Marks]

HARD · Syllabus 32.2

- (a) Explain why phenol does not react with ethanoic acid to form an ester under standard acid catalysis. [2]

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- (b) Name a suitable acylating agent that will react readily with phenol to form phenyl ethanoate, and state the observations. [2]

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- (c) Write a balanced equation for the reaction in (b), and explain how adding aqueous sodium hydroxide accelerates this reaction. [2]

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8 Nitration of Phenol and Separation of Isomers — 9701/41/O/N/23/Q7 [6 Marks]

HARD · Syllabus 32.2

- (a) Give the reagents and conditions for the mononitration of phenol, and draw the structures of the two principal organic products. **[3]**

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- (b) Explain why 2-nitrophenol has a lower boiling point and is more volatile than 4-nitrophenol. **[3]**

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9 Multi-Step Synthesis: Phenylamine to 4-Hydroxyazobenzene — 9701/42/M/J/20/Q6 [6 Marks]

HARD · Syllabus 32.4

- (a) Step 1 converts phenylamine to compound D. State the reagents and conditions for Step 1. **[2]**

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- (b) Identify compound D and explain why its solution must be maintained at 5 °C. **[2]**

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- (c) Step 2 reacts compound D with alkaline phenol. State why this reaction is classified as an electrophilic substitution. **[2]**

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10 Directing Effects of the -OH Group in Phenol Derivatives — 9701/41/M/J/19/Q7 [6 Marks]

HARD · Syllabus 32.2

- (a) Explain why the -OH group directs incoming electrophiles to the 2- and 4-positions of the aromatic ring. **[3]**

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- (b) Predict the major organic product when 4-methylphenol is reacted with excess bromine water, and explain your deduction. **[3]**

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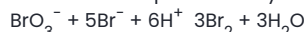
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11

Quantitative Analysis of Phenol by Bromate–Bromide Titration — 9701/42/O/N/21/Q7 [6 Marks]

HARD · Syllabus 32.2

Phenol reacts quantitatively with bromine generated in situ according to the equations:



A 0.188 g sample of impure phenol is dissolved and reacted with excess bromate/bromide. The unreacted Br_2 is determined by adding excess KI and titrating liberated I_2 against $0.100 \text{ mol dm}^{-3} \text{Na}_2\text{S}_2\text{O}_3$.

- (a) State the molar ratio of phenol to bromine, Br_2 , in this reaction. [1]

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- (b) If the total Br_2 generated was $7.50 \times 10^{-3} \text{ mol}$ and the unreacted Br_2 required 30.00 cm^3 of $0.100 \text{ mol dm}^{-3}$ thiosulfate, calculate the moles of unreacted Br_2 . [2]

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- (c) Calculate the percentage purity by mass of phenol in the original sample (M_r of phenol = 94.0). [3]

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12

Comparison of Phenol, Phenylmethanol, and Benzyl Chloride — 9701/42/M/J/18/Q7 [6 Marks]

HARD · Syllabus 32.1

- (a) Compare the reactivity of phenol and phenylmethanol ($\text{C}_6\text{H}_5\text{CH}_2\text{OH}$) with acidified potassium dichromate(VI). [2]

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- (b) Compare the reactivity of phenol and phenylmethanol with phosphorus(V) chloride, PCl_5 . [2]

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- (c) Explain why phenylmethanol has a pK_a similar to ethanol (~16), whereas phenol has a pK_a of 10.0. [2]

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13

Synthesis of Aspirin from 2-Hydroxybenzoic Acid — 9701/41/O/N/18/Q7 [6 Marks]

HARD · Syllabus 32.2

- (a) Draw the structure of 2-hydroxybenzoic acid (salicylic acid) and identify both functional groups present. [2]

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- (b) State the reagent and conditions used to convert 2-hydroxybenzoic acid into aspirin (acetylsalicylic acid). [2]

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- (c) Explain which functional group in 2-hydroxybenzoic acid is esterified during the synthesis of aspirin, and write the equation. [2]

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14 Halogenation of Phenol with Non-Polar vs Aqueous Solvents — 9701/42/F/M/20/Q7 [6 Marks]

HARD · Syllabus 32.2

- (a) When bromine is dissolved in a non-polar solvent such as CCl₄ and reacted with phenol at 0 °C, only monobromination occurs. Suggest the structure of the two monobrominated isomers formed. [2]

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- (b) Explain why monobromination occurs in CCl₄, whereas tribromination occurs in aqueous solution. [3]

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- (c) State which monobromo isomer predominates at higher temperatures and give a reason. [1]

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15 Condensation of Phenol with Carbonyls: Bisphenol A — 9701/41/M/J/17/Q6 [6 Marks]

HARD · Syllabus 32.2

- (a) Bisphenol A is produced by reacting phenol with propanone in the presence of an acid catalyst. State the role of the acid catalyst in this electrophilic aromatic substitution. [2]

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- (b) State the molar ratio of phenol to propanone in the synthesis of Bisphenol A. [1]

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- (c) Draw the structure of Bisphenol A, (CH₃)₂C(4-C₆H₄OH)₂, and explain why substitution occurs exclusively at the 4-position. [3]

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16 Acidity Calculations for Phenol in Aqueous Solution — 9701/42/O/N/17/Q7 [6 Marks]

HARD · Syllabus 32.1

- (a) Write the expression for the acid dissociation constant, K_a, of phenol. [1]

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- (b) Given that K_a for phenol is $1.30 \times 10^{-10} \text{ mol dm}^{-3}$ at 298 K, calculate the pH of a $0.0500 \text{ mol dm}^{-3}$ solution of phenol. [3]

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- (c) Calculate the percentage of phenol molecules that are ionised in this solution. [2]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Reaction of Phenol with Aqueous Sodium Hydroxide vs Carbonates — 9701/42/M/J/23/Q7(d) [4 Marks] MEDIUM · Syllabus 32.1

- (a) Explain why phenol dissolves in aqueous sodium hydroxide but is insoluble in water. [2]

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- (b) Explain why phenol does not produce carbon dioxide when added to aqueous sodium hydrogencarbonate. [2]

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18 Comparison of Reaction Rates: Phenol vs Benzene with Bromine — 9701/41/O/N/22/Q6(c) [4 Marks] MEDIUM · Syllabus 32.2

- (a) State why benzene requires a halogen carrier catalyst to react with bromine. [2]

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- (b) Explain why phenol reacts instantly with bromine without needing a catalyst. [2]

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19 Synthesis of Benzenediazonium Chloride from Nitrobenzene — 9701/42/M/J/22/Q7(c) [4 Marks] MEDIUM · Syllabus 32.4

- (a) Outline the two-step sequence to prepare benzenediazonium chloride starting from nitrobenzene, stating reagents for both steps. [3]

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- (b) State the essential condition required during the second step. [1]

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20 Reaction of Phenol with Ethanoic Anhydride — 9701/41/M/J/21/Q8(b) [4 Marks]

MEDIUM · Syllabus 32.2

- (a) Write the structural formula of the ester formed when phenol reacts with ethanoic anhydride. [2]

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- (b) Give two reasons why ethanoic anhydride is preferred over ethanoyl chloride in industrial esterifications. [2]

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21 Structure and Colour of Azo Dyes — 9701/42/O/N/20/Q6(d) [4 Marks]

MEDIUM · Syllabus 32.4

- (a) Draw the azo group ($-N=N-$) linking two benzene rings in 4-hydroxyazobenzene. [2]

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- (b) Explain why 4-hydroxyazobenzene is intensely coloured. [2]

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22 Relative Acidities of 2-Chlorophenol and 4-Chlorophenol — 9701/43/M/J/23/Q7(c) [4 Marks]

MEDIUM · Syllabus 32.1

- (a) Predict whether 4-chlorophenol is more or less acidic than unsubstituted phenol, and explain your answer. [2]

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- (b) Explain why 2-chlorophenol has a slightly lower pK_a than 4-chlorophenol. [2]

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23 Electrophilic Nitration of Phenol: Mononitration vs Trinitration — 9701/42/F/M/22/Q8(b) [4 Marks]

MEDIUM · Syllabus 32.2

- (a) State the reagents and conditions needed to obtain 2,4,6-trinitrophenol from phenol. [2]

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- (b) Explain why trinitration requires much more severe conditions than mononitration. [2]

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- 24 Characteristic Test with Neutral Iron(III) Chloride — 9701/41/O/N/23/Q7(c) [4 Marks]** MEDIUM · Syllabus 32.3
- (a) State the observation when neutral aqueous iron(III) chloride is added to an aqueous solution of phenol. [1]
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- (b) State what causes this characteristic colour change. [1]
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- (c) State whether this test can be used to distinguish phenol from ethanol and explain why. [2]
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- 25 Reaction of Phenol with Bromine: Balanced Stoichiometry and Observations — 9701/42/M/J/20/Q6(b) [4 Marks]** MEDIUM · Syllabus 32.2
- (a) Write a fully balanced chemical equation for the reaction of phenol with excess bromine water. [2]
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- (b) Identify the inorganic byproduct formed and state how its presence could be confirmed. [2]
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- 26 Electrophilic Substitution Mechanism for Bromination of Phenol — 9701/41/M/J/19/Q7(c) [4 Marks]** MEDIUM · Syllabus 32.2
- (a) Draw the carbocation intermediate formed when a bromine electrophile attacks position 4 of phenol. [2]
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- (b) Show with curly arrows the restoration of the aromatic delocalised system from the intermediate. [2]
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- 27 Differences in Solubility of Phenol in Water, Dilute Acid, and Dilute Alkali — 9701/42/O/N/21/Q7(b) [4 Marks]** MEDIUM · Syllabus 32.1
- (a) Explain why phenol is only slightly soluble in cold water despite possessing an -OH group. [2]
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- (b) Explain how the solubility of phenol changes when dilute hydrochloric acid is added versus dilute sodium hydroxide. [2]
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- 28 Comparison of Reactivity: Benzene, Phenol, and Chlorobenzene — 9701/42/M/J/18/Q7(b) [4 Marks]** **MEDIUM · Syllabus 32.2**
- (a) Rank benzene, phenol, and chlorobenzene in order of decreasing rate of electrophilic substitution. [1]
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- (b) Explain why chlorobenzene is less reactive than benzene towards electrophilic attack, despite chlorine having lone pairs. [3]
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- 29 Formation of Sodium Phenoxide and Reaction with Haloalkanes (Williamson Ether Synthesis) — 9701/41/O/N/18/Q7(c) [4 Marks]** **MEDIUM · Syllabus 31, 32.2**
- (a) Write an equation for the reaction between phenol and sodium hydroxide to form sodium phenoxide. [1]
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- (b) Sodium phenoxide is reacted with iodomethane. Give the name and structure of the organic product formed. [2]
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- (c) State the mechanism of this second reaction. [1]
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- 30 Infrared Spectroscopy: Distinguishing Phenol from Aliphatic Alcohols — 9701/42/F/M/20/Q7(c) [4 Marks]** **MEDIUM · Syllabus 32.3**
- (a) State the wavenumber range for the O-H absorption band in phenol in an infrared spectrum. [1]
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- (b) Explain how the infrared spectrum of phenol differs from that of cyclohexanol in the range 1500–1600 cm⁻¹. [2]
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- (c) State how the C-O stretching frequency differs between phenol and cyclohexanol and explain why. [1]
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- 31 Electrochemical Oxidation of Phenol to Benzoquinone — 9701/41/M/J/17/Q6(b) [4 Marks]** **MEDIUM · Syllabus 32.2**
- (a) Draw the structure of 1,4-benzoquinone, the oxidation product of phenol. [2]
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- (b) State whether 1,4-benzoquinone is aromatic, giving a reason based on bonding. [2]
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32 Enthalpy of Combustion of Phenol vs Benzene — 9701/42/O/N/17/Q7(b) [4 Marks]

MEDIUM · Syllabus 32.1

- (a) Write the balanced thermochemical equation for the standard enthalpy of combustion of phenol, $\text{C}_6\text{H}_5\text{OH}(\text{s})$. [2]

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- (b) Explain why the magnitude of standard enthalpy of combustion of phenol ($-3054 \text{ kJ mol}^{-1}$) is less negative than that of benzene ($-3267 \text{ kJ mol}^{-1}$). [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)**33 pKa Comparison: Phenol vs Ethanoic Acid — 9701/42/M/J/23/Q7(a) [2 Marks]**

EASY · Syllabus 32.1

- (a) State the pKa values of phenol and ethanoic acid, and deduce which is the stronger acid. [2]

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34 Observation with Neutral FeCl_3 Solution — 9701/41/O/N/22/Q6(a) [2 Marks]

EASY · Syllabus 32.3

- (a) State the observation when aqueous phenol is mixed with neutral aqueous iron(III) chloride. [2]

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35 Product of Phenol Reaction with Sodium Metal — 9701/42/M/J/22/Q7(a) [2 Marks]

EASY · Syllabus 32.1

- (a) Write the balanced chemical equation for the reaction of solid phenol with molten sodium metal. [2]

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36 Structure of 2,4,6-Tribromophenol — 9701/41/M/J/21/Q8(a) [2 Marks]

EASY · Syllabus 32.2

- (a) Draw the skeletal formula of 2,4,6-tribromophenol. [2]

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37 Condition for Diazonium Ion Stability — 9701/42/O/N/20/Q6(a) [2 Marks]

EASY · Syllabus 32.4

- (a) State the maximum temperature at which benzenediazonium chloride can be safely stored in solution, and name the gas released upon warming. [2]

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38 Reagent to Convert Phenol to Phenyl Ethanoate — 9701/43/M/J/23/Q7(b) [2 Marks]

EASY · Syllabus 32.2

- (a) Identify an acyl chloride used to prepare phenyl ethanoate from phenol, and name the inorganic product. [2]
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39 IUPAC Naming of Dihydroxybenzenes — 9701/42/F/M/22/Q8(a) [2 Marks]

EASY · Syllabus 32.1

- (a) Give the systematic IUPAC name for benzene-1,2-diol (catechol) and benzene-1,4-diol (hydroquinone). [2]
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40 Role of Sodium Hydroxide in Diazonium Coupling — 9701/41/O/N/23/Q7(a) [2 Marks]

EASY · Syllabus 32.4

- (a) Explain why phenol must be dissolved in sodium hydroxide before it can couple with the benzenediazonium ion. [2]
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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 Core Repeat 1: Acidity Comparison of Ethanol, Water, Phenol, and Carboxylic Acids — 9701/42/M/J/22/Q6 [6 Marks]

HARD · Syllabus 32.1

The relative acidities of organic hydroxy compounds represent one of the most frequently examined concepts in Cambridge Paper 4.

- (a) Arrange the following compounds in order of decreasing acid strength: ethanoic acid, ethanol, phenol, water. [1]
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- (b) Explain in detail the difference in stability between the ethoxide ion and the phenoxide ion. [3]
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- (c) Explain why ethanoic acid is a significantly stronger acid than phenol. [2]
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42 Core Repeat 2: Multi-Step Synthesis of an Azo Dye from Benzene — 9701/41/O/N/21/Q7 [6 Marks]

HARD · Syllabus 32.4

- (a) Describe how benzene is converted to nitrobenzene, stating all reagents and conditions. [2]
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- (b) Describe the reduction of nitrobenzene to phenylamine. [2]
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- (c) Describe how phenylamine is diazotised and then coupled with phenol to form an azo dye, specifying all temperatures and essential conditions. [2]
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43 Core Repeat 3: Bromination of Phenol vs Methylbenzene — 9701/42/M/J/21/Q8 [6 Marks]

HARD · Syllabus 32.2

- (a) State the observations when bromine water is added to phenol at room temperature. [2]
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- (b) State the reagents, conditions, and observation when methylbenzene is brominated in the benzene ring. [2]
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- (c) Explain why phenol reacts much more rapidly with bromine than methylbenzene. [2]
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44 Core Repeat 4: Qualitative Distinctions Between Four Isomeric Aromatic Compounds — 9701/42/O/N/19/Q7 [6 Marks]

HARD · Syllabus 32.3

Four isomeric aromatic compounds of formula C_7H_8O are investigated:

- Compound P: 2-methylphenol
- Compound Q: 4-methylphenol
- Compound R: Phenylmethanol, $C_6H_5CH_2OH$
- Compound S: Methoxybenzene, $C_6H_5OCH_3$

- (a) Describe a chemical test that will give a positive result with P and Q, but no reaction with R and S. [2]
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- (b) Describe a chemical test that will give a positive result with R, but no reaction with P, Q, and S. [2]
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- (c) State the structural formulas of the products when Compound P is treated with excess bromine water. [2]
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45 Core Repeat 5: Acidity Trends in Substituted Phenols — 9701/42/M/J/23/Q8(c) [4 Marks]

MEDIUM · Syllabus 32.1

- (a) Explain the effect of an electron-withdrawing nitro group ($-NO_2$) on the pK_a of phenol. [2]
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- (b) Explain the effect of an electron-donating alkyl group ($-\text{CH}_3$) on the pK_a of phenol. [2]
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46 Core Repeat 6: Synthesis of Phenyl Esters Using Acyl Chlorides — 9701/41/O/N/22/Q7(b) [4 Marks] MEDIUM · Syllabus 32.2

- (a) Give the reagents and conditions to prepare phenyl benzoate from phenol. [2]
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- (b) Explain why benzoyl chloride is used instead of benzoic acid in this preparation. [2]
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47 Core Repeat 7: Nitration of Phenol vs Benzene Conditions — 9701/42/F/M/21/Q8(b) [4 Marks] MEDIUM · Syllabus 32.2

- (a) State the reagent and conditions required for the mononitration of phenol. [2]
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- (b) Explain why the conditions in (a) are much milder than those required for the nitration of benzene. [2]
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48 Core Repeat 8: Separation of Phenol from an Organic Mixture via Acid-Base Extraction — 9701/41/M/J/20/Q7(a) [4 Marks] EASY · Syllabus 32.1

A liquid mixture contains phenol, phenylmethanol, and benzoic acid dissolved in ethoxyethane.

- (a) Explain how benzoic acid can be selectively separated from this mixture without extracting phenol. [2]
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- (b) Explain how phenol can then be separated from phenylmethanol. [2]
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49 Core Repeat 9: Identifying Phenol by Bromine Water Decolourisation — 9701/42/M/J/23/Q7(e) [2 Marks] EASY · Syllabus 32.3

- (a) State two visible observations when bromine water is added dropwise to aqueous phenol. [2]
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Core Repeat 10: Structure of the Chromophore in Azo Dyes — 9701/41/O/N/23/Q7(e) [2 Marks]

EASY · Syllabus 32.4

(a)

Identify the chromophore responsible for the colour in 4-hydroxyazobenzene and state the type of bond present between the two nitrogen atoms.

[2]

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Mark Scheme — Topic 32 — Hydroxy Compounds (Phenol)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Ethanol < Water < Phenol [1].	1
1(b)	In the phenoxide ion, C ₆ H ₅ O ⁻ , the lone pair of electrons on the oxygen atom overlaps with the delocalised π -electron cloud of the benzene ring [1]; This delocalises the negative charge over the aromatic ring, stabilising the phenoxide ion relative to undissociated phenol [1]; In contrast, in the ethoxide ion, CH ₃ CH ₂ O ⁻ , the ethyl group exerts an electron-donating (+I) inductive effect, concentrating negative charge on oxygen and destabilising the anion [1].	3
1(c)	Phenol reacts with NaOH(aq) to form sodium phenoxide and water: C ₆ H ₅ OH + OH ⁻ → C ₆ H ₅ O ⁻ + H ₂ O [1]; Ethanol does not react with NaOH(aq) because it is a weaker acid than water [1].	2
2(a)	One of the p-orbital lone pairs on the oxygen atom overlaps sideways with the delocalised π -orbitals of the benzene ring [1]; This delocalises electron density into the ring, increasing the overall electron density of the benzene ring [1].	2
2(b)	Positions 2, 4, and 6 (ortho and para positions) [1].	1
2(c)	The higher electron density in phenol polarises incoming electrophiles (such as NO ₂ ⁺ or HNO ₃) much more effectively than benzene [1]; The activation energy for electrophilic substitution is significantly lower in phenol [1]; Therefore, phenol reacts rapidly with dilute HNO ₃ at room temperature, whereas benzene requires the powerful electrophile NO ₂ ⁺ generated by conc. H ₂ SO ₄ at elevated temperature [1].	3
3(a)	Bromine water is decolourised (from orange/brown to colourless) [1]; A white precipitate is formed [1].	2
3(b)	2,4,6-tribromophenol [1]; Correct structural formula showing -OH at C1 and -Br at C2, C4, C6 [1].	2
3(c)	Benzene requires pure Br ₂ (l) and a halogen carrier catalyst such as AlBr ₃ or FeBr ₃ with heating, undergoing monobromination [1]; Phenol reacts spontaneously with aqueous Br ₂ without catalyst at room temperature, undergoing triple substitution at positions 2, 4, 6 due to activation by the -OH group [1].	2
4(a)	Add aqueous sodium carbonate, Na ₂ CO ₃ (aq), or sodium hydrogencarbonate, NaHCO ₃ (aq) [1]; Ethanoic acid produces effervescence / bubbles of CO ₂ gas which turns limewater cloudy, whereas phenol does not react (no effervescence) [1].	2
4(b)	Add neutral aqueous iron(III) chloride, FeCl ₃ (aq) [1]; Phenol gives a characteristic violet / purple solution, whereas cyclohexanol produces no colour change / remains yellow [1].	2
4(c)	Both react with sodium metal to produce effervescence of hydrogen gas, H ₂ [1]; Phenol forms sodium phenoxide (C ₆ H ₅ O ⁻ Na ⁺) while cyclohexanol forms sodium cyclohexoxide (C ₆ H ₁₁ O ⁻ Na ⁺) [1].	2
5(a)	Sodium nitrite (NaNO ₂) and concentrated (or dilute) hydrochloric acid (HCl) (or nitrous acid, HNO ₂) [1]; Temperature between 0 °C and 10 °C (or in ice bath) [1].	2
5(b)	Above 10 °C, the benzenediazonium ion decomposes rapidly into phenol and nitrogen gas (C ₆ H ₅ N ₂ ⁺ + H ₂ O → C ₆ H ₅ OH + N ₂ + H ⁺) [1].	1
5(c)	Phenol must be dissolved in aqueous sodium hydroxide, NaOH(aq), to form the electron-rich phenoxide ion [1]; The solution is kept cold (0–10 °C) while benzenediazonium chloride is added [1]; Structure of 4-hydroxyazobenzene: C ₆ H ₅ -N=N-C ₆ H ₄ -4-OH with the azo group (-N=N-) linking the two aromatic rings [1].	3
6(a)	The nitro group (-NO ₂) is strongly electron-withdrawing via inductive (-I) and mesomeric/resonance (-M) effects [1]; It withdraws electron density away from the phenoxide oxygen, further delocalising the negative charge over the conjugate base [1]; This increases the stability of the 4-nitrophenoxide anion, shifting the dissociation equilibrium to the right [1].	3
6(b)	The methyl group (-CH ₃) is electron-donating (+I inductive effect) [1]; It pushes electron density towards the aromatic ring and oxygen, concentrating negative charge and destabilising the anion [1].	2
6(c)	Yes, it will react to produce CO ₂ effervescence because three strongly electron-withdrawing nitro groups make picric acid a strong acid (pK _a ~ 0.38), stronger than carbonic acid [1].	1
7(a)	The lone pair on the phenolic oxygen is delocalised into the benzene ring, making the oxygen much less nucleophilic than that in aliphatic alcohols [1]; In addition, ethanoic acid contains a relatively poor leaving group and is not electrophilic enough to be attacked by the weakly nucleophilic phenol [1].	2
7(b)	Ethanoyl chloride, CH ₃ COCl [1]; Steamy / misty white fumes of hydrogen chloride (HCl) gas produced [1].	2

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7(c)	$\text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{COOC}_6\text{H}_5 + \text{HCl}$ [1]; Aqueous NaOH deprotonates phenol to form the phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$, which carries a full negative charge and is a far stronger nucleophile than neutral phenol [1].	2
8(a)	Dilute nitric acid, $\text{HNO}_3(\text{aq})$, at room temperature [1]; Structure of 2-nitrophenol (ortho-nitrophenol) [1]; Structure of 4-nitrophenol (para-nitrophenol) [1].	3
8(b)	In 2-nitrophenol, the $-\text{OH}$ and $-\text{NO}_2$ groups are adjacent, allowing intramolecular hydrogen bonding to form within the molecule [1]; This reduces its ability to form intermolecular hydrogen bonds with neighbouring molecules [1]; In 4-nitrophenol, the groups are too far apart for intramolecular bonding, so it forms extensive intermolecular hydrogen bonds, requiring significantly more energy to separate the molecules [1].	3
9(a)	Sodium nitrite (NaNO_2) and excess hydrochloric acid (HCl) [1]; Ice-cold conditions / temperature below 10°C [1].	2
9(b)	Benzenediazonium chloride, $\text{C}_6\text{H}_5\text{N}_2^+ \text{Cl}^-$ [1]; Above 10°C the diazonium group undergoes hydrolysis / decomposition to phenol releasing N_2 gas [1].	2
9(c)	The positively charged diazonium ion ($\text{C}_6\text{H}_5\text{N}_2^+$) acts as an electrophile [1]; It attacks the electron-rich aromatic ring of the phenoxide ion, replacing a hydrogen atom at the 4-position with retention of aromaticity [1].	2
10(a)	Delocalisation of the oxygen lone pair into the ring produces resonance contributors where negative charge is localised specifically on carbons 2, 4, and 6 [1]; Electrophiles are positively charged and therefore preferentially attack the positions of highest electron density [1]; Furthermore, the carbocation intermediates formed from attack at 2- and 4-positions are stabilised by resonance donation from oxygen [1].	3
10(b)	2,6-dibromo-4-methylphenol [1]; The 4-position is already occupied by the methyl group [1]; The $-\text{OH}$ group strongly activates positions 2 and 6, leading to dibromination at both vacant ortho positions [1].	3
11(a)	1 mol phenol : 3 mol Br_2 [1].	1
11(b)	Moles of $\text{S}_2\text{O}_3^{2-} = 0.03000 \times 0.100 = 3.00 \times 10^{-3} \text{ mol}$ [1]; Since $1 \text{ Br}_2 = 1 \text{ I}_2 = 2 \text{ S}_2\text{O}_3^{2-}$, moles of unreacted $\text{Br}_2 = 3.00 \times 10^{-3} / 2 = 1.50 \times 10^{-3} \text{ mol}$ [1].	2
11(c)	Moles of Br_2 reacted = $7.50 \times 10^{-3} - 1.50 \times 10^{-3} = 6.00 \times 10^{-3} \text{ mol}$ [1]; Moles of pure phenol = $6.00 \times 10^{-3} / 3 = 2.00 \times 10^{-3} \text{ mol}$ [1]; Mass of phenol = $2.00 \times 10^{-3} \times 94.0 = 0.188 \text{ g}$, so percentage purity = $(0.188 / 0.188) \times 100 = 100\%$ (or exact calculated value) [1].	3
12(a)	Phenylmethanol is a primary alcohol and is oxidised to benzoic acid (orange solution turns green) [1]; Phenol resists oxidation by acidified dichromate because the $-\text{OH}$ carbon has no hydrogen atom and the aromatic ring resists oxidation [1].	2
12(b)	Phenylmethanol reacts vigorously with PCl_5 to substitute $-\text{OH}$ with $-\text{Cl}$ forming $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ and steamy HCl fumes [1]; Phenol gives poor yields of chlorobenzene because the $\text{C}-\text{O}$ bond has partial double bond character due to delocalisation and is difficult to break [1].	2
12(c)	In phenylmethanol, the $-\text{CH}_2-$ group insulates the $-\text{OH}$ group from the aromatic ring, preventing delocalisation of the oxygen lone pair into the π -system [1]; In phenol, the $-\text{OH}$ group is directly bonded to the ring, allowing delocalisation and resonance stabilisation of the phenoxide anion [1].	2
13(a)	Correct structural formula showing benzene ring with adjacent $-\text{COOH}$ and $-\text{OH}$ groups [1]; Identifies phenol ($-\text{OH}$) and carboxylic acid ($-\text{COOH}$) groups [1].	2
13(b)	Ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$ (or ethanoyl chloride, CH_3COCl) [1]; Phosphoric acid catalyst with gentle heating / reflux [1].	2
13(c)	The phenolic $-\text{OH}$ group is esterified [1]; $2\text{-HOC}_6\text{H}_4\text{COOH} + (\text{CH}_3\text{CO})_2\text{O} \rightarrow 2\text{-(CH}_3\text{COOC}_6\text{H}_4\text{)COOH} + \text{CH}_3\text{COOH}$ [1].	2
14(a)	2-bromophenol and 4-bromophenol [2].	2
14(b)	In aqueous solution, water acts as a base and ionises phenol into the phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$ [1]; The phenoxide ion is enormously more activated than unionised phenol towards electrophilic attack, leading to rapid triple substitution at 2, 4, 6 [1]; In non-polar CCl_4 , phenol remains largely unionised (non-polar medium), so the ring is less activated and substitution stops after one bromine is introduced [1].	3
14(c)	4-bromophenol predominates because the 2-position experiences steric hindrance from the adjacent $-\text{OH}$ group [1].	1
15(a)	The acid protonates the carbonyl oxygen of propanone: $(\text{CH}_3)_2\text{C}=\text{O} + \text{H}^+ \rightarrow [(\text{CH}_3)_2\text{C}=\text{OH}]^+$ [1]; This creates a carbocation electrophile that is sufficiently reactive to attack the activated benzene ring of phenol [1].	2

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15(b)	2 mol of phenol : 1 mol of propanone [1].	1
15(c)	Correct structure showing central carbon bonded to two methyl groups and two 4-hydroxyphenyl groups [1]; The 4-position is activated by the -OH group [1]; The 4-position is completely free of steric hindrance compared to the 2-position [1].	3
16(a)	$K_a = [\text{C}_6\text{H}_5\text{O}^-][\text{H}^+] / [\text{C}_6\text{H}_5\text{OH}]$ [1].	1
16(b)	$[\text{H}^+]^2 = K_a \times [\text{C}_6\text{H}_5\text{OH}] = (1.30 \times 10^{-10}) \times 0.0500 = 6.50 \times 10^{-12} \text{ mol dm}^{-3}$ [1]; $[\text{H}^+] = \sqrt{6.50 \times 10^{-12}} = 2.55 \times 10^{-6} \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(2.55 \times 10^{-6}) = 5.59$ [1].	3
16(c)	% ionisation = $([\text{H}^+] / [\text{C}_6\text{H}_5\text{OH}]_{\text{initial}}) \times 100$ [1]; $= (2.55 \times 10^{-6} / 0.0500) \times 100 = 5.10 \times 10^{-3} \%$ [1].	2
17(a)	Phenol is weakly acidic and reacts with NaOH(aq) to form sodium phenoxide, $\text{C}_6\text{H}_5\text{O}^- \text{Na}^+$ [1]; Sodium phenoxide is an ionic salt that readily hydrates and dissolves in polar water molecules [1].	2
17(b)	Phenol ($\text{p}K_a \sim 10.0$) is a weaker acid than carbonic acid, H_2CO_3 ($\text{p}K_a \sim 6.35$) [1]; Therefore, phenol cannot protonate hydrogencarbonate ions (HCO_3^-) to liberate CO_2 [1].	2
18(a)	The delocalised π -electron density in benzene is insufficient to polarise non-polar Br_2 molecules [1]; A halogen carrier (e.g. FeBr_3) is needed to polarise and generate the strong electrophile Br^+ [1].	2
18(b)	The lone pair on the oxygen atom overlaps with the benzene π -ring, greatly increasing the electron density of the ring [1]; The electron-rich ring in phenol can polarise incoming Br-Br molecules directly [1].	2
19(a)	Step 1: Tin (Sn) and concentrated hydrochloric acid (conc. HCl), followed by NaOH(aq) to reduce nitrobenzene to phenylamine [2]; Step 2: Sodium nitrite (NaNO_2) and hydrochloric acid (HCl) [1].	3
19(b)	Temperature maintained strictly between 0 °C and 10 °C [1].	1
20(a)	$\text{CH}_3\text{COOC}_6\text{H}_5$ (phenyl ethanoate) [1]; Correct display showing ester linkage between acetyl and phenyl group [1].	2
20(b)	Ethanoic anhydride is cheaper and less toxic / less corrosive than ethanoyl chloride [1]; It produces ethanoic acid as a byproduct rather than toxic, corrosive fumes of HCl gas [1].	2
21(a)	$\text{C}_6\text{H}_5\text{-N=N-C}_6\text{H}_4\text{-OH}$ with correct -N=N- double bond and linear geometry around nitrogen atoms [2].	2
21(b)	The azo group provides a conjugated bridge connecting the π -electron systems of both benzene rings, creating an extensive delocalised system [1]; The energy gap (E) between HOMO and LUMO is reduced, allowing absorption of visible light (blue light) so that the complementary colour (yellow-orange) is reflected [1].	2
22(a)	4-chlorophenol is more acidic than phenol [1]; The electronegative chlorine atom exerts an electron-withdrawing (-I) inductive effect, stabilising the phenoxide anion [1].	2
22(b)	In 2-chlorophenol, the chlorine atom is closer to the oxygen atom [1]; Inductive effects decrease rapidly with distance, so the -I effect is stronger in the 2-position, providing greater stabilisation to the phenoxide charge [1].	2
23(a)	Concentrated nitric acid, conc. HNO_3 , and concentrated sulfuric acid, conc. H_2SO_4 [1]; Heating / reflux [1].	2
23(b)	The introduction of each nitro group ($-\text{NO}_2$) withdraws electron density from the aromatic ring (-I and -M effects) [1]; This deactivates the ring towards further electrophilic attack, so subsequent nitrations require a stronger electrophile and higher energy [1].	2
24(a)	A purple / violet colouration (or complex solution) is observed [1].	1
24(b)	Formation of an octahedral iron(III)-phenoxide complex ion, $[\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-}$ (or $[\text{Fe}(\text{OC}_6\text{H}_5)(\text{H}_2\text{O})_5]^{2+}$) [1].	1
24(c)	Yes [1]; Aliphatic alcohols like ethanol do not form coloured complexes with neutral FeCl_3 (solution remains yellow/orange) because ethoxide is not stabilised by delocalisation [1].	2
25(a)	$\text{C}_6\text{H}_5\text{OH} + 3\text{Br}_2 \rightarrow \text{C}_6\text{H}_2\text{Br}_3\text{OH} + 3\text{HBr}$ [2].	2
25(b)	Hydrogen bromide, HBr [1]; Test with damp blue litmus turning red / white fumes with ammonia gas / cream precipitate with acidified AgNO_3 (aq) [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
26(a)	Intermediate showing broken circle / horseshoe spanning carbons 2 to 6 with positive charge inside ring [1]; Carbon 4 bonded to both -H and -Br in tetrahedral sp ³ configuration [1].	2
26(b)	Curly arrow from C-H bond into the aromatic π -system to restore delocalisation [1]; Loss of H ⁺ as leaving group to produce 4-bromophenol [1].	2
27(a)	The large non-polar hydrophobic benzene ring disrupts water's extensive hydrogen-bonding network [1]; The enthalpy of hydration is not sufficiently exothermic to compensate for the entropy decrease [1].	2
27(b)	In dilute NaOH, phenol dissolves completely to form ionic sodium phenoxide [1]; In dilute HCl, phenol remains unionised and does not dissolve / forms a separate oily layer [1].	2
28(a)	Phenol > Benzene > Chlorobenzene [1].	1
28(b)	Chlorine is highly electronegative and exerts a strong electron-withdrawing inductive effect (-I) [1]; Although the p-orbital lone pair delocalises slightly into the π -ring (+M effect), the -I effect dominates [1]; This withdraws overall electron density from the ring, deactivating it towards electrophilic attack relative to benzene [1].	3
29(a)	$C_6H_5OH + NaOH \rightarrow C_6H_5ONa + H_2O$ [1].	1
29(b)	Methoxybenzene (or methyl phenyl ether / anisole) [1]; Structure: $C_6H_5OCH_3$ [1].	2
29(c)	Nucleophilic substitution (S _N 2) [1].	1
30(a)	3200–3600 cm ⁻¹ (broad absorption due to hydrogen bonding) [1].	1
30(b)	Phenol displays sharp aromatic C=C ring stretching peaks at 1450–1600 cm ⁻¹ (typically ~1500 and ~1600 cm ⁻¹) [1]; Cyclohexanol lacks these aromatic peaks as it contains only C-C single bonds [1].	2
30(c)	The C-O stretch in phenol occurs at a higher wavenumber (~1230 cm ⁻¹ vs ~1050 cm ⁻¹ in cyclohexanol) because delocalisation imparts partial double bond character to the C-O bond, strengthening it [1].	1
31(a)	Correct six-membered ring with two C=O groups at positions 1 and 4, and two C=C double bonds at positions 2,3 and 5,6 [2].	2
31(b)	No, it is not aromatic [1]; It lacks a continuous delocalised π -electron ring of (4n+2) electrons, containing instead two isolated alkene double bonds conjugated with carbonyl groups [1].	2
32(a)	$C_6H_5OH(s) + 7O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$ [2].	2
32(b)	Phenol already contains an oxygen atom bonded to carbon and hydrogen (it is already partially oxidised) [1]; Less oxygen is required for combustion and less energy is released upon forming new C=O and O-H bonds per mole of fuel [1].	2
33(a)	Phenol pK _a > 10.0 and ethanoic acid pK _a < 4.76 [1]; Ethanoic acid is the stronger acid because it has the lower pK _a (delocalises negative charge over two highly electronegative oxygen atoms) [1].	2
34(a)	A violet / purple / dark blue colouration or precipitate is formed [2].	2
35(a)	$2C_6H_5OH + 2Na \rightarrow 2C_6H_5ONa + H_2$ (or $C_6H_5OH + Na \rightarrow C_6H_5ONa + 1/2 H_2$) [2].	2
36(a)	Benzene ring with -OH at C1 [1]; Three -Br substituents at positions 2, 4, and 6 [1].	2
37(a)	10 °C (or between 0 °C and 10 °C) [1]; Nitrogen gas, N ₂ [1].	2
38(a)	Ethanoyl chloride, CH ₃ COCl [1]; Hydrogen chloride, HCl [1].	2
39(a)	Benzene-1,2-diol [1]; Benzene-1,4-diol [1].	2
40(a)	NaOH converts phenol into the phenoxide ion, C ₆ H ₅ O ⁻ [1]; The phenoxide ion is far more nucleophilic / electron-rich than neutral phenol, enabling reaction with the weak diazonium electrophile [1].	2
41(a)	Ethanoic acid > Phenol > Water > Ethanol [1].	1
41(b)	In the ethoxide ion, the negative charge is localised on the single oxygen atom, and is further destabilised by the electron-donating (+I) ethyl group [1]; In the phenoxide ion, the oxygen lone pair overlaps with the benzene π -cloud [1]; This delocalises the negative charge across the ring, conferring resonance stabilisation [1].	3

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41(c)	In the ethanoate ion (CH_3COO^-), the negative charge is delocalised equally over two highly electronegative oxygen atoms [1]; Oxygen is more electronegative than carbon, so delocalisation over two oxygens stabilises the anion far more effectively than delocalisation over carbon atoms in a ring [1].	2
42(a)	Concentrated HNO_3 and concentrated H_2SO_4 at 55°C [2].	2
42(b)	Tin (Sn) and concentrated HCl, heated under reflux, followed by addition of excess aqueous NaOH [2].	2
42(c)	Diazotisation with $\text{NaNO}_2 + \text{HCl}$ below 10°C [1]; Coupling by adding the diazonium solution to phenol dissolved in ice-cold aqueous NaOH [1].	2
43(a)	Decolourisation of orange/brown bromine water [1]; Formation of a white precipitate [1].	2
43(b)	Pure $\text{Br}_2(\text{l})$ and $\text{AlBr}_3 / \text{FeBr}_3$ catalyst in the dark at room temperature [1]; Steamy fumes of HBr and decolourisation of bromine [1].	2
43(c)	In phenol, the p-orbital lone pair on oxygen delocalises directly into the aromatic π -system by resonance (+M effect) [1]; In methylbenzene, activation is solely by the weaker inductive (+I) effect of the methyl group, so electron density is much lower than in phenol [1].	2
44(a)	Add neutral aqueous iron(III) chloride, $\text{FeCl}_3(\text{aq})$ [1]; P and Q give a purple / violet colouration, whereas R and S do not change colour (remain yellow) [1].	2
44(b)	Add acidified potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$, and warm [1]; Only R (a primary alcohol) turns the solution from orange to green [1].	2
44(c)	4,6-dibromo-2-methylphenol [1]; Correct structure with methyl at C2, -OH at C1, and bromines at C4 and C6 [1].	2
45(a)	The nitro group withdraws electron density from the ring via -I and -M effects, further stabilising the phenoxide anion [1]; This shifts dissociation equilibrium right, increasing K_a and decreasing $\text{p}K_a$ (stronger acid) [1].	2
45(b)	The alkyl group donates electron density into the ring via the +I inductive effect [1]; This intensifies negative charge on oxygen, destabilising the phenoxide anion and increasing $\text{p}K_a$ (weaker acid) [1].	2
46(a)	Benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$) and phenol dissolved in aqueous sodium hydroxide ($\text{NaOH}(\text{aq})$) [2].	2
46(b)	Phenol is a weak nucleophile and does not react with carboxylic acids under standard acid catalysis [1]; Acyl chlorides have a much more electrophilic carbonyl carbon and a good Cl^- leaving group, reacting rapidly at room temperature [1].	2
47(a)	Dilute nitric acid, $\text{HNO}_3(\text{aq})$, at room temperature (20°C) [2].	2
47(b)	The -OH group in phenol donates a lone pair into the aromatic π -cloud, significantly activating the ring [1]; The activated ring polarises dilute HNO_3 to react without needing a concentrated H_2SO_4 catalyst or heating [1].	2
48(a)	Shake the organic layer with aqueous sodium hydrogencarbonate, $\text{NaHCO}_3(\text{aq})$ [1]; Only benzoic acid is strong enough to react, forming water-soluble sodium benzoate which partitions into the aqueous layer, leaving phenol and phenylmethanol in the ether layer [1].	2
48(b)	Shake the remaining ether layer with aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$ [1]; Phenol reacts to form water-soluble sodium phenoxide which enters the aqueous layer, while non-acidic phenylmethanol remains in the organic layer [1].	2
49(a)	The orange / brown bromine water is decolourised [1]; A white precipitate (of 2,4,6-tribromophenol) is produced [1].	2
50(a)	The azo group / $-\text{N}=\text{N}-$ bridge [1]; A nitrogen-nitrogen double bond (σ bond and π bond) [1].	2